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ASSIGNMENT 1

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Review Concepts (Problems 1–12)

Look up, as necessary, the following review concepts. Then, after waiting a while, write the mathematical definitions from memory, and give a real-world example. Then for each, check your work, and revise as necessary.

Problem 1

Random variable.

SOLUTION:

In the real world, a random variable is a numerical quantity whose value is settled by the outcome of a random phenomenon—a measurement that has not yet been made. The mathematics mirrors the measuring procedure exactly: given a sample space S , a random variable is a function

$$X : S \rightarrow \mathbb{R}$$

assigning to each possible outcome $s \in S$ a number $X(s)$. The function itself is deterministic; the randomness lives entirely in which outcome s the experiment produces. Random variables are written with capital letters, their observed values with lower case (Problem 8).

The standard first example already contains the whole idea. Flip a coin twice and let X be the number of heads: $S = \{HH, HT, TH, TT\}$, with $X(HH) = 2$, $X(HT) = X(TH) = 1$, and $X(TT) = 0$, as drawn in Figure 1. A measurement procedure is precisely a rule turning whatever happened into a number, which is why the function formulation is the natural one; that two outcomes share the value 1 is not a defect of the picture but the origin of $P(X = 1) = \frac{1}{2}$.

The literature is unanimous about the object and divided only over ceremony. Ross introduces it in passing—random variables are “these real-valued functions defined on the sample space” (§4.1, p. 112)—where DeGroot and Schervish give the same sentence the status of a numbered definition (Def. 3.1.1, p. 93), and Blitzstein and Hwang (Def. 3.1.1, p. 104) surround it with cautions against the circular gloss “a variable that takes on random values.” Pitman postpones the formality altogether: he works with distributions for two full chapters before random variables enter informally (§3.1, p. 139), the function-on- Ω definition surviving only as a technical remark (pp. 143–144). HPS does not define the term at all; its first page speaks of “the observed values of a random variable X ,” the Preface having referred all probability background to the companion Volume I. This assignment, in other words, reconstructs exactly the material the course text takes as given. In a fully rigorous treatment X must also be measurable, so that $P(X \in B)$ is defined for the sets one cares about; no book at this level imposes the condition, and nothing in this course requires it, so the point is recorded here once and suppressed below.

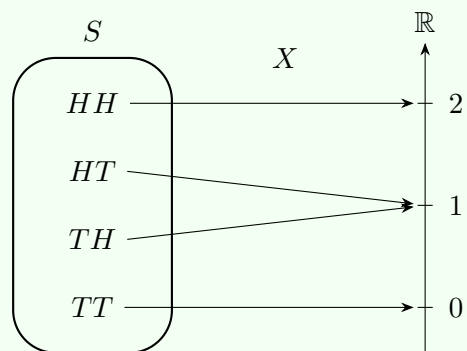


Figure 1. A RANDOM VARIABLE IS A FUNCTION. The arrows are fixed by the definition of X ; the experiment chooses only the row. Two outcomes landing on 1 is the origin of $P(X = 1) = \frac{1}{2}$.



Problem 2

Sample space.

SOLUTION:

The sample space is, in real-world terms, an exhaustive list of everything that could happen in the random phenomenon being modeled. Mathematically, it is the set

$$S = \{\text{all possible outcomes of the experiment}\},$$

exactly one element of which occurs when the experiment is run. Events are subsets $A \subseteq S$, and random variables are functions defined on S . Writing down S is itself a modeling decision: the outcomes must be mutually exclusive and collectively exhaustive, and the description must be fine enough that every quantity of interest is a well-defined function on S .

Ross's formulation is the canonical one: "This set of all possible outcomes of an experiment is known as the sample space of the experiment and is denoted by S " (§2.2, p. 21). For two coin flips, $S = \{HH, HT, TH, TT\}$; for the waiting time, in minutes, for the next 1 train at 116th Street, $S = [0, \infty)$. The two-flip space distinguishes HT from TH even though both yield one head, and that finer description is what makes the heads count of Problem 1 a well-defined function on S .

Beneath the shared definition there are genuine differences of scope. Ross admits every subset of S as an event (p. 22), while DeGroot and Schervish require the events to satisfy three closure conditions— S itself is an event, complements of events are events, countable unions of events are events (pp. 6–10)—which amounts to a σ -field in all but name. The caution is not pedantry: in uncountable spaces some sets cannot

consistently carry a probability at all, a point taken up under Problem 5. Pitman declines the term itself, writing *outcome space* throughout and explaining why in his preface: the space holds outcomes, not samples (§1.1, p. 2; p. vi). His objection lands with particular force in this course, because HPS uses “sample space” in the statistician’s sense, for the n -dimensional space of possible sample *values*—the eight vertices of a unit cube, for instance, when three coin tosses are recorded as a binary vector (pp. 3, 53). When the term appears in these notes it will usually carry HPS’s meaning: the range of the data vector rather than an abstract Ω . ■

Problem 3

Distribution.

SOLUTION:

Before a random variable is observed, everything there is to know about it is how likely it is to land anywhere; that complete description is its distribution. Mathematically, the distribution of X is the assignment

$$A \mapsto P(X \in A)$$

of a probability to each set $A \subseteq \mathbb{R}$. It is determined by the cumulative distribution function $F(x) = P(X \leq x)$, and in practice it is handled through a probability mass function when X is discrete and a density when X is continuous. Distinct random variables can share one distribution: the number of heads and the number of tails in two fair flips are different functions on S , yet both are distributed as $\text{Binomial}(2, \frac{1}{2})$.

A fair die roll has the distribution assigning $1/6$ to each of $1, 2, \dots, 6$. The heads count of Problem 1 has $P(X = 0) = P(X = 2) = \frac{1}{4}$ and $P(X = 1) = \frac{1}{2}$; Figure 2 gives its two standard portraits, and the redundancy between them is instructive—the jump of F at x equals $P(X = x)$, so either panel reconstructs the other.

The texts differ mainly over which description to treat as primary. Ross leads with the cdf and its characterizing properties, nondecreasing, right-continuous, with limits 0 and 1 (§4.1, p. 116; §4.10, pp. 159–160), on the view that “all probability questions about X can be answered in terms of the c.d.f.” DeGroot and Schervish instead define the distribution as the full collection of probabilities $\Pr(X \in C)$, concede that the definition is “awkward,” and develop the cdf and densities as its working summaries

(Def. 3.1.2, p. 94); their passing remark that a distribution is itself a probability measure on the real line is, in more advanced language, the pushforward construction $B \mapsto P(X^{-1}(B))$. Pitman builds in the opposite order, defining distributions as mass spread over a space before random variables exist and only later specializing to the law of X (§1.3, pp. 19–21; §3.1, pp. 140–141). Blitzstein and Hwang contribute the warning this problem quietly turns on: a distribution is a “blueprint” (p. 128), and conflating a random variable with its blueprint—what they call sympathetic magic (pp. 127–128)—is exactly the confusion the heads-and-tails example above refutes. HPS assumes the concept outright and puts it to work on its first page, posing that “the distribution of X is assumed to belong to a certain family of distributions”: the parametric families this course will estimate and test.

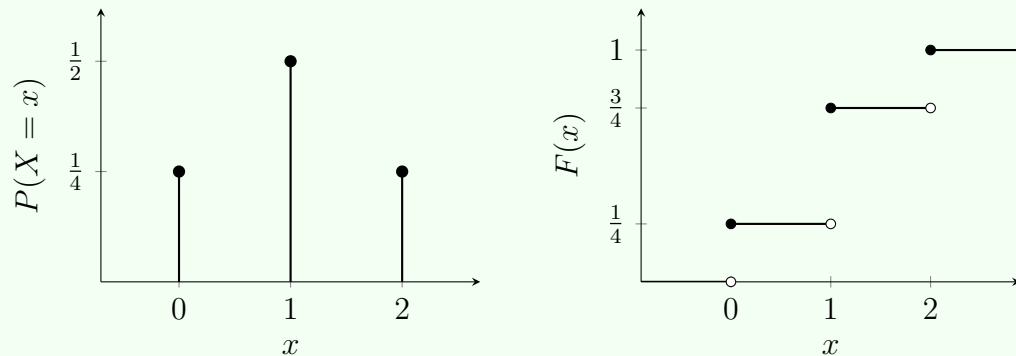


Figure 2. TWO PORTRAITS OF ONE DISTRIBUTION. The pmf (left) and cdf (right) of the heads count $X \sim \text{Binomial}(2, \frac{1}{2})$. The jumps of F are the pmf values, so either panel reconstructs the other.



Problem 4

Joint distribution.

SOLUTION:

Several random variables measured on the same experiment behave as a package, and the joint distribution is the complete description of that package, dependence included. For $X_1, \dots, X_n$ defined on a common sample space, the joint distribution is the assignment

$$B \mapsto P((X_1, \dots, X_n) \in B), \quad B \subseteq \mathbb{R}^n.$$

It determines every marginal distribution (constrain one coordinate and leave the rest free), but the marginals do not determine it. If X_1, X_2 are indicators of heads for two different fair coins and X is the indicator for a single fair coin, then (X_1, X_2) and (X, X) have identical Bernoulli($\frac{1}{2}$) marginals, yet the first spreads probability $\frac{1}{4}$ over four outcomes while the second concentrates everything on the diagonal.

That gap between marginals and joint is the operative fact, not a technicality. Sample a Columbia student and record height X and weight Y : the marginals describe how heights vary and how weights vary, and say nothing about whether tall students tend to be heavy—the co-movement on which any study of the relationship depends lives in the joint distribution alone. Sklar's theorem gives the gap its modern form: every joint cdf factors into its marginals and a copula, a pure dependence function, so specifying a joint law amounts to specifying the marginals and the dependence separately.

Ross constructs the object from the joint cdf, $F(a, b) = P\{X \leq a, Y \leq b\}$ (§6.1, p. 220), recovering the marginals by sending the other argument to infinity—a maneuver that quietly invokes the continuity of probability. DeGroot and Schervish define it

as the collection of all probabilities $\Pr[(X, Y) \in C]$ (Def. 3.4.1, p. 118; n variables in §3.7), and are alone in treating mixed pairs with one discrete and one continuous coordinate. Blitzstein and Hwang open their chapter with the motivation: the individual pmfs are each a complete blueprint for one variable and still “missing important information” about how the two move together (pp. 303–304). Pitman keeps the discrete case concrete—the joint distribution is the table of $P(x, y)$, which he insists on orienting like the coordinate plane rather than like a matrix (pp. 144–145)—and describes the continuous case as a probability distribution over the plane (p. 339). For HPS one joint distribution matters above all others: that of the whole sample, which under random sampling factors as $\prod_{i=1}^n f(x_i | \theta)$ (§1.2, p. 6). That product is the likelihood, and much of this course is the study of it. ■

Problem 5

Axioms of probability.

SOLUTION:

In real-world terms, the axioms say that probability behaves like one unit of belief—or of long-run frequency—split consistently over all the ways things can turn out. Mathematically, a probability P on a sample space S assigns each event $A \subseteq S$ a number $P(A)$ subject to three requirements:

$$P(A) \geq 0, \quad P(S) = 1, \quad P\left(\bigcup_i A_i\right) = \sum_i P(A_i) \text{ for disjoint } A_1, A_2, \dots$$

—nonnegativity, normalization, and (countable) additivity, the last holding in particular for finitely many disjoint events. Everything else in the calculus of probability is a consequence: $P(A^c) = 1 - P(A)$, $P(\emptyset) = 0$, monotonicity, inclusion–exclusion. A fair die wears all three axioms on its face: each side carries probability $\frac{1}{6} \geq 0$, the six sides exhaust what can happen and their probabilities total 1, and additivity over disjoint faces gives $P(\text{even}) = P(\{2\}) + P(\{4\}) + P(\{6\}) = \frac{1}{2}$.

It is a curious bibliographical fact that the books do not agree even on how many axioms there are, though the disagreements are matters of packaging rather than substance. Ross posts three, folding the upper bound into the first, $0 \leq P(E) \leq 1$ (p. 26). DeGroot and Schervish also count three, with countable additivity third and finite additivity derived from it as a theorem (Thm. 1.5.2, pp. 16–18); they take care to present the axioms as neutral ground among the frequency, classical, and subjective interpretations surveyed in their §1.2, a neutrality of evident importance to DeGroot, a committed subjectivist. Blitzstein and Hwang compress the list to

two, packaging $P(\emptyset) = 0$ with $P(S) = 1$ (Def. 1.6.1, p. 22) and deriving the rest (Thm. 1.6.2, p. 23). Pitman avoids the word “axiom” entirely: probability obeys the same “rules of proportion” that govern area, mass, and percentages of a population (§1.3, p. 21), and countable additivity is deferred until it is needed, as the infinite sum rule of §3.4 (p. 209). HPS states no axioms at all— $P(\cdot)$ first appears on p. 34 already computing a normal probability—the foundations having been settled in Volume I. All of these descend from Kolmogorov’s 1933 axiomatization. Two technical remarks complete the picture: countable additivity is equivalent to the continuity of P along monotone sequences of events, which is what the limit theorems of probability run on; and the restriction of “event” to a σ -field of subsets, rather than all subsets, is forced in uncountable spaces, where the axiom of choice produces sets—Vitali’s construction is the classical example—that no suitably invariant probability can measure consistently.

■

Problem 6

Joint probability mass function.

SOLUTION:

For discrete random variables, the joint probability mass function gives the probability of every completely specified combination of values. Mathematically, the joint pmf of $X_1, \dots, X_n$ is

$$p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n),$$

with $p \geq 0$ and $\sum_{x_1, \dots, x_n} p(x_1, \dots, x_n) = 1$. The probability of any event is obtained by summing p over the combinations in the event, and marginal pmfs come from summing the other coordinates out: $p_{X_1}(x) = \sum_{x_2, \dots, x_n} p(x, x_2, \dots, x_n)$.

Two flips of a fair coin, with X_1, X_2 the indicators of heads, give $p(x_1, x_2) = \frac{1}{4}$ for each of the four pairs in $\{0, 1\}^2$. Summing out the second coordinate recovers the marginal, $p_{X_1}(1) = p(1, 0) + p(1, 1) = \frac{1}{2}$, and summing over the event that the flips agree gives $p(0, 0) + p(1, 1) = \frac{1}{2}$ —the two basic operations, marginalization and event probability, both instances of the same addition rule.

The object is completely standard; its name is not. Ross and Blitzstein and Hwang both say *joint probability mass function* (§6.1, p. 221; Def. 7.1.2, p. 304, with the latter’s Figure 7.1 drawing it as bars over the plane with total height one). DeGroot and Schervish pointedly do not: “We will not use that term again in this text,” they write of “p.m.f.” (Def. 3.1.4, p. 96), preferring *joint p.f.* (Def. 3.4.3, p. 119). Pitman dispenses with a name altogether, working from the display of joint probabilities in a table oriented like the coordinate plane; the row and column sums of that table sit literally in its margins, which he notes is why marginal distributions are called

marginal (pp. 144–145). HPS dissolves the discrete/continuous divide by convention, announcing on p. 2 that “the word density is used for both discrete and continuous random variables.” The convention is sounder than it may first appear: a pmf is a density with respect to counting measure rather than length, so a single word—and a single set of formulas—legitimately covers both cases. The Bernoulli likelihood $\theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}$ of their §2.4 (p. 25) is this convention at work. ■

Problem 7

Joint density.

SOLUTION:

Continuous random variables put probability zero on every exact point, so their joint behavior is described by a concentration—probability per unit volume—rather than by point masses. A function $f \geq 0$ is the joint density of $X_1, \dots, X_n$ when

$$P((X_1, \dots, X_n) \in B) = \int_B f(x_1, \dots, x_n) dx_1 \cdots dx_n \quad \text{for all } B \subseteq \mathbb{R}^n,$$

with $\int_{\mathbb{R}^n} f = 1$. Only regions carry probability. A density value is not a probability and can exceed one; the reliable reading is always $f(x) dx \approx P(x \leq X \leq x + dx)$. Marginal densities come from integrating the other variables out.

The four probability texts define the same object with different first moves. Ross posits the existence of an f satisfying the integral identity (Eq. 1.3, p. 223) and immediately supplies the infinitesimal reading $P\{a < X < a + da, b < Y < b + db\} \approx f(a, b) da db$ (p. 224); Pitman leads with that infinitesimal form, $P(X \in dx, Y \in dy) = f(x, y) dx dy$, “the probability per unit area for values near (x, y) ” (p. 349), after introducing the density surface whose volumes over regions are probabilities (p. 346). DeGroot and Schervish require the integral identity over every subset of the plane (Def. 3.4.4, pp. 120–121), while Blitzstein and Hwang define the joint pdf as the mixed partial derivative of the joint cdf (Def. 7.1.13, p. 312) and repeat the caution stated above: points, lines, and entire curves all carry probability zero. In the advanced formulation the density is a Radon–Nikodym derivative of the distribution with respect to Lebesgue measure—the same role the pmf plays with respect to counting measure,

so the unification behind HPS’s “density for both” convention (Problem 6) extends here as well. As for HPS itself, the joint density of consequence is the sample’s: equation (2) of §4.1 (p. 113) is the normal regression likelihood $\prod_i f(y_i | x_i)$, precisely the joint density the linear-model portion of this course will differentiate and maximize.

Sibling heights make the concept concrete: the joint density of a randomly sampled pair piles up near the diagonal of $\mathbb{R}^2$, since taller-than-average people tend to have taller-than-average siblings. Figure 3 sketches the level curves of such a density. And for a density exceeding one: if (X, Y) is uniform on the square $[0, \frac{1}{2}]^2$, then $f \equiv 4$ there, legitimately, since $4 \times \frac{1}{4} = 1$.

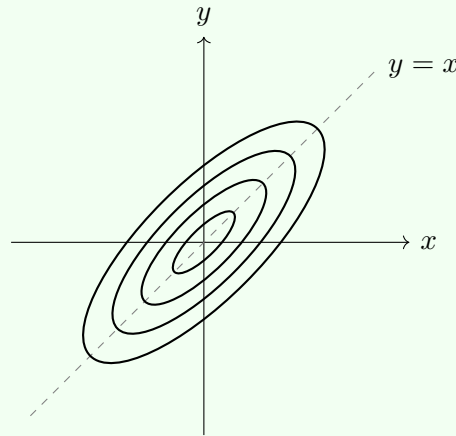


Figure 3. LEVEL CURVES OF A JOINT DENSITY. A positively dependent pair such as sibling heights, centered at the means. Each ellipse is a set where f is constant; the probability of a region is the volume under the density surface above it, and the mass concentrates along $y = x$.



Problem 8

Realizations.

SOLUTION:

A realization is the value actually observed on one run of the experiment—the data point in hand, as opposed to the not-yet-observed random variable. Mathematically, if the experiment produced the outcome $s \in S$, the realization of the random variable X is the number

$$x = X(s),$$

and the notational convention keeps the two straight: capital letters for random variables, lower case for realizations. A die about to be rolled is a random variable; the 4 it shows is a realization; tomorrow’s roll is a fresh realization of a variable with the *same* distribution. Data sets are realizations. This course designs methods as functions of random variables, so that they have distributions and risks, and applies them to realizations, so that they produce numbers.

On this item, unusually, the course text is the most explicit of the five. HPS legislates the convention on p. 2—“a capital letter normally represents a random variable and the corresponding small letter its numerical value”—and immediately builds on it the distinction the course runs on: an *estimator* is “a function of the random variables, whereas the word estimate denotes its numerical value” (p. 8), a split enforced throughout the book (N_i against n_i on p. 93, Y_{ij} against y_{ij} on p. 129). The other authors leave more unsaid. Ross records that $X(s)$ denotes “the value of X when $s \in S$ is the outcome of the experiment” (§4.9, p. 155) but never announces the case convention; DeGroot and Schervish observe it consistently under the name “observed

value,” the word “realization” occurring exactly once in their eight hundred pages. Blitzstein and Hwang supply the sharpest guardrail: $X = x$ denotes an *event*, the set $\{s : X(s) = x\}$ —which is why an expression like “ $P(X)$ ” is meaningless—and once the outcome is realized, the random variable “crystallizes” into the number $X(s)$ (pp. 106–107). Pitman reads the lowercase letter differently again, as a bound dummy variable ranging over possible values in sums such as $P(X = k) = \binom{4}{k} 2^{-4}$ (p. 141): the algebraic role of x rather than its temporal one. The framings differ; the discipline they enforce is one and the same. ■

Problem 9

Variance.

SOLUTION:

The variance measures how spread out a random quantity is around its expectation—the size of the typical squared miss. For a random variable Y with $\mu = E\{Y\}$ (and $E\{Y^2\}$ finite),

$$\text{Var}\{Y\} = E\{(Y - \mu)^2\} = E\{Y^2\} - (E\{Y\})^2,$$

the second form following by expansion and linearity: $E\{Y^2 - 2\mu Y + \mu^2\} = E\{Y^2\} - 2\mu^2 + \mu^2$. Variance is nonnegative, equals zero exactly when Y is constant with probability one, and carries squared units; its square root, the standard deviation σ , lives on the scale of Y itself. It is not linear: added constants drop out entirely and scale factors emerge squared, $\text{Var}\{aY + c\} = a^2 \text{Var}\{Y\}$, a rule Problem 14 exercises.

The mathematics is uniform across the texts; the motivations are not. Ross and Blitzstein and Hwang both justify squaring by the deficiencies of the absolute deviation $E|X - \mu|$, which Ross calls “mathematically inconvenient” (§4.5, p. 126; Def. 4.6.1 and Thm. 4.6.2, pp. 171–172). DeGroot and Schervish attend to existence—no finite mean, no variance (Def. 4.3.1, p. 226)—and promote the basic facts to theorems, including that zero variance characterizes a point mass (Thms. 4.3.2–4.3.3, p. 229). Pitman titles his section for the standard deviation rather than the variance and says plainly why both notions persist: “Variance is harder to interpret than SD, but has simpler algebraic properties” (§3.3, p. 185); his verbal form of the shortcut—the mean of the square minus the square of the mean (p. 186)—is worth carrying around. Most

consequential for this course is how HPS arrives at the concept: not as descriptive spread but as risk, “the mean squared error $E(d - \theta)^2$ becomes the variance of d when $E(d) = \theta$ ” (§2.1, p. 15). Variance is what squared-error risk collapses to for unbiased estimators, and it will shadow every estimator we meet. Two structural facts round this out. The map $c \mapsto E\{(Y - c)^2\}$ is minimized at $c = \mu$, with minimum value $\text{Var}\{Y\}$ —add and subtract μ and the cross term vanishes—which is the bias–variance decomposition in embryo. And Chebyshev’s inequality, $P(|Y - \mu| \geq k\sigma) \leq 1/k^2$, converts the variance from a definition into an operational guarantee about tails.

The concept is visible in any pair of measuring instruments. Two thermometers may both read the true temperature on average, the cheap one scattering about $\pm 2^\circ$ and the lab-grade one about $\pm 0.2^\circ$: identical expectations, variances apart by a factor of a hundred. Figure 4 plots the two densities; buying the better instrument is buying lower variance.

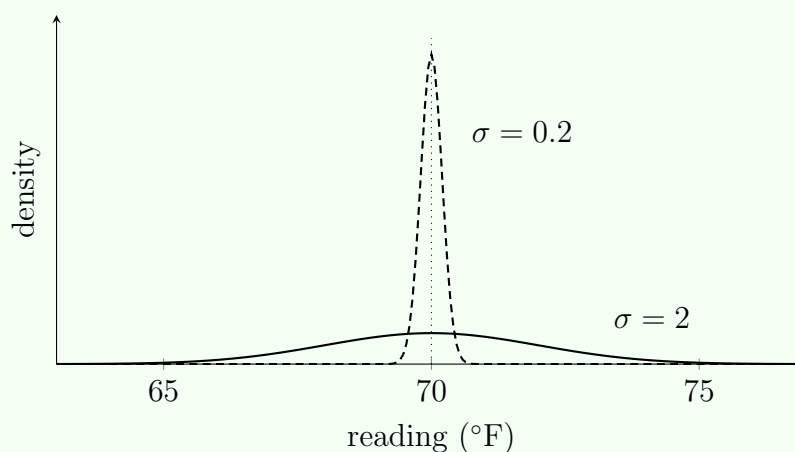


Figure 4. EQUAL MEANS, UNEQUAL VARIANCES. Both thermometer densities are centered at the true temperature $\mu = 70$, so both instruments are unbiased; the entire difference in quality is variance.



Problem 10

Covariance.

SOLUTION:

The covariance measures whether two random quantities tend to miss their expectations in the same direction, in opposite directions, or in no linearly patterned way at all. For random variables X and Y with expectations μ_X, μ_Y ,

$$\text{Cov}(X, Y) = E\{(X - \mu_X)(Y - \mu_Y)\} = E\{XY\} - E\{X\}E\{Y\},$$

the shortcut form again by expansion and linearity. Covariance is symmetric, unaffected by added constants, bilinear— $\text{Cov}(aU + bV, W) = a \text{Cov}(U, W) + b \text{Cov}(V, W)$, and likewise in the second slot—and satisfies $\text{Cov}(Y, Y) = \text{Var}\{Y\}$. Height and weight across a population covary positively; outdoor temperature and heating cost, negatively; shoe size and grade-point average, not at all, so far as anyone can tell. Figure 5 shows the three regimes as data. Because the magnitude carries the product of the two units, one standardizes to the correlation $\rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y) \in [-1, 1]$ when comparability matters.

The definition is identical in all four probability texts (Ross §7.4, p. 305; DeGroot and Schervish Def. 4.6.1, p. 248; Blitzstein and Hwang Def. 7.3.1, p. 326; Pitman p. 430); the differences lie in the role each assigns it. Pitman is the most explicit about why the concept exists at all: covariance “appears in calculation of the variance of a sum of possibly dependent random variables”—it is the cross term—and he defers it until that need arises, then notes the identity $\text{Cov}(I_A, I_B) = P(AB) - P(A)P(B)$ (p. 431), under which covariance generalizes the very definition of independent events.

Blitzstein and Hwang name the boundary case, calling variables with zero covariance *uncorrelated* (p. 327), and prove the correlation bounds (Thm. 7.3.5, p. 330); DeGroot and Schervish obtain existence from finite variances via the Schwarz inequality (§4.6). HPS recalls covariance from Volume I exactly twice, and both recalls point at this course’s destination: once to build the correlation coefficient and its sample version (§2.6, p. 33), and once assembled into the covariance matrix $[\sigma_{ij}] = E(Y - \mu)(Y - \mu)'$ of Theorem 3 (§4.5, p. 125), the object behind the least-squares result $E(\hat{\beta} - \beta)(\hat{\beta} - \beta)' = \sigma^2(X'X)^{-1}$ (p. 126). A useful way to organize all the algebra: on centered square-integrable variables, covariance is an inner product, variance the squared norm, and correlation the cosine of the angle; the bound $|\rho| \leq 1$ is the Cauchy–Schwarz inequality, and bilinearity—the engine of Problem 15—is simply what inner products do.

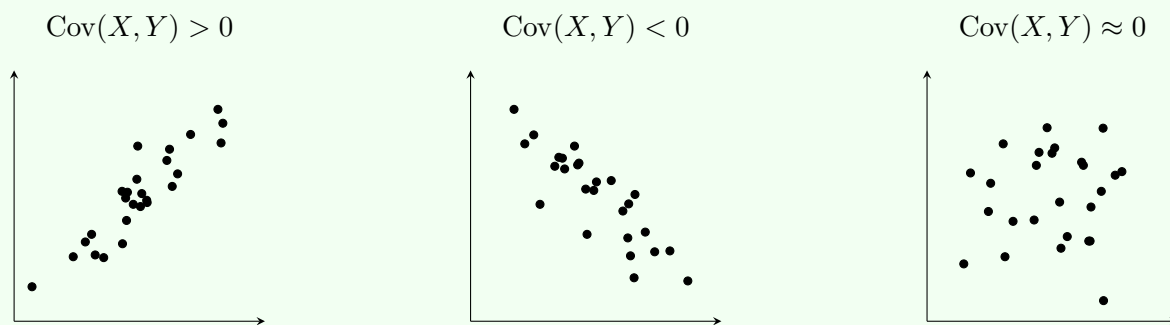


Figure 5. THE SIGN OF COVARIANCE. Simulated clouds of 26 points each. Left: deviations from the means agree in direction. Center: they oppose. Right: no linear pattern; this cloud’s sample correlation is +0.06.

■

Problem 11

Independence between random variables.

SOLUTION:

Random variables are independent when learning the value of one tells you nothing about the others. Mathematically, X and Y are independent when their joint distribution factors into the product of the marginals:

$$P(X \in A, Y \in B) = P(X \in A) P(Y \in B) \quad \text{for all } A, B \subseteq \mathbb{R};$$

equivalently, the joint pmf or density factors, $p(x, y) = p_X(x)p_Y(y)$ for all (x, y) —Figure 6 shows the geometry of that factorization. The colloquial reading is literal: whenever $P(Y \in B) > 0$, the definition yields $P(X \in A \mid Y \in B) = P(X \in A)$, so conditioning moves nothing. For several variables, mutual independence requires the joint to factor into the product of all n marginals for every choice of sets. In a clinical trial, the outcomes of two different patients are modeled this way: one patient’s recovery tells you nothing about another’s, although both outcomes inform the unknown recovery *rate*—the independence is a statement within the model, for each fixed value of that rate.

Ross (Eq. 2.1, p. 228) and DeGroot and Schervish (Def. 3.5.2, p. 135) take the event factorization above as the definition and derive the working criteria as theorems: the joint cdf factors, the joint pmf factors, the joint density factors—and, in a refinement worth remembering, it suffices for the joint density to factor into *any* product $h_1(x)h_2(y)$, marginal or not (DeGroot and Schervish, Thm. 3.5.5, pp. 135–136). Blitzstein and Hwang begin from the factoring cdf (Def. 3.8.1, p. 129) and add two facts this course

will lean on: functions of independent random variables are independent (Thm. 3.8.5, p. 130), and independent is not the same as identically distributed, a distinction they lay out as a two-by-two table (box 3.8.7). Pitman contributes the equivalence closest to the intuition, that the conditional distributions of Y given $X = x$ do not vary with x (p. 152, with the discrete product rule on p. 151 and the general event form in §4.1, p. 261). HPS, consistent with its policy, never defines independence: it acquires it through the definition of random sampling—the observations “will be independently and identically distributed” (§1.2, pp. 5–6)—and puts the closure property to serious work at p. 151, where functions of independent normal variables deliver the independence of $\bar{X}$ and the sample variance, a fact the book concedes was “used without proof on several occasions in the earlier chapters.” One refinement deserves precision: mutual independence is strictly stronger than pairwise independence. Take X and Y independent fair-coin indicators and let Z indicate that they agree; each pair among X, Y, Z is independent, yet the triple is not, since any two of them determine the third.

$p_Y(1) = \frac{1}{3}$	$\frac{1}{2} \cdot \frac{1}{3}$	$\frac{1}{2} \cdot \frac{1}{3}$
$p_Y(0) = \frac{2}{3}$	$\frac{1}{2} \cdot \frac{2}{3}$	$\frac{1}{2} \cdot \frac{2}{3}$
	$p_X(0) = \frac{1}{2}$	$p_X(1) = \frac{1}{2}$

Figure 6. INDEPENDENCE AS A PRODUCT GRID. For independent $X \sim \text{Bernoulli}(\frac{1}{2})$ and $Y \sim \text{Bernoulli}(\frac{1}{3})$, each joint probability is the area of a cell: row total times column total. A joint pmf whose table is not such a product grid is dependent.



Problem 12

The implication of independence for covariance.

SOLUTION:

If X and Y are independent (with finite variances), their covariance is zero. The proof takes two lines: independence makes expectations of products split, $E\{XY\} = E\{X\}E\{Y\}$ —factor the joint pmf or density and sum or integrate—so

$$\boxed{\text{Cov}(X, Y) = E\{XY\} - E\{X\}E\{Y\} = E\{X\}E\{Y\} - E\{X\}E\{Y\} = 0}.$$

Two coins flipped in different rooms are the live instance: independent, hence uncorrelated; Problem 15 uses precisely this implication to eliminate a term. The converse fails, and the standard counterexample repays writing out. Let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Then $E\{X\} = 0$ and $E\{XY\} = E\{X^3\} = \frac{1}{3}(-1 + 0 + 1) = 0$, so $\text{Cov}(X, Y) = 0$; yet Y is a function of X —dependence could not be stronger, as $P(Y = 1 \mid X = 1) = 1$ while $P(Y = 1) = \frac{2}{3}$. Figure 7 shows why covariance is blind here: the dependence is pure curvature, and covariance registers only linear association. The implication is strictly one-directional.

Every text proves the implication and immediately cautions against its converse; the counterexamples they choose are more interesting than the theorem. Ross takes X uniform on $\{-1, 0, 1\}$ with Y indicating the event $X = 0$, so that $XY \equiv 0$ (Prop. 4.1 of Ch. 7 and the following remark, pp. 304–305). DeGroot and Schervish and Pitman both use the parabola example above—the former as Example 4.6.4, “Dependent but Uncorrelated Random Variables” (pp. 251–252), with the uniform distribution on the unit disk as a continuous second witness; the latter as Exercise 6.4.5 (p. 445), the

moral having been set in a box: “Independent variables are uncorrelated, but uncorrelated variables are not necessarily independent” (p. 433). Blitzstein and Hwang run the same construction through $X \sim N(0, 1)$, $Y = X^2$, where the odd moments of the normal vanish by symmetry (Thm. 7.3.2, p. 327), and draw the conclusion that matters: covariance measures *linear* association. One family does redeem the converse—if (X, Y) is *jointly* normal, zero covariance implies independence—and the adverb carries the theorem, since marginal normality is not enough: for $Y = SX$ with S a random sign independent of $X \sim N(0, 1)$, both marginals are standard normal and $\text{Cov}(X, Y) = E\{S\} E\{X^2\} = 0$, yet $|Y| = |X|$. HPS never states any of this, and nevertheless exploits the distinction more pointedly than the books that do: Theorem 3 of §4.5 (p. 125) assumes only *uncorrelated* errors ($\sigma_{ij} = 0$ for $i \neq j$) to deliver the means and variances of the least-squares estimators, while the distributional conclusions one page later require “that we are willing to assume that the Y ’s are independent normal variables” (p. 126). The space between what uncorrelatedness buys and what independence buys is where the linear-model chapters of this course will operate.

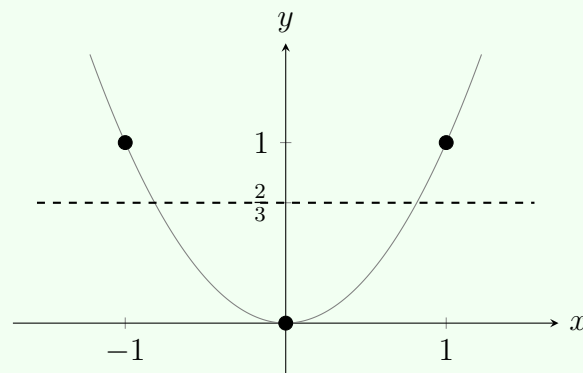


Figure 7. DEPENDENT BUT UNCORRELATED. All probability lies on the parabola $y = x^2$ (three equal atoms), so Y is a function of X ; the best-fitting line (dashed, at height $E\{Y\} = \frac{2}{3}$) is nonetheless flat. Covariance registers only linear association.

■

Moment Computations (Problems 13–15)

Suppose that you have three random variables, X , Y , and Z with expectations, respectively, 1, 2, and 3, variances, respectively, 4, 9, 16, and suppose that X and Y are independent and that the covariance between X and Z is 1 and the covariance between Y and Z is -1 . Work out the following, showing all your steps, and stating the rule used in each step.

Problem 13

Work out $E\{2X + Y + 7\}$.

SOLUTION:

$$\begin{aligned} E\{2X + Y + 7\} &= E\{2X\} + E\{Y\} + E\{7\} && \text{(sum rule)} \\ &= 2E\{X\} + E\{Y\} + 7 && \text{(scaling rule; constant rule)} \\ &= 2(1) + 2 + 7 && \text{(substitute the givens)} \\ &= \boxed{11}. \end{aligned}$$

The first step is the sum rule: $E\{U + V\} = E\{U\} + E\{V\}$, with no assumption about how U and V are related. The texts are emphatic on that clause—“it holds even if X and Y are dependent” (Blitzstein and Hwang, Thm. 4.2.1, p. 152); additivity holds “regardless of what their joint distribution is” (DeGroot and Schervish, Thm. 4.2.4, p. 219); the addition rule applies “no matter whether X and Y are independent or not” (Pitman, p. 167). The second step applies the scaling and constant rules, $E\{aU\} = aE\{U\}$ and $E\{c\} = c$ (Ross, Cor. 4.1 of Ch. 4, p. 125; DeGroot and Schervish,

Cor. 4.2.1, p. 218). Together the three rules constitute the linearity of expectation, whose closed form $E\{aX + bY + c\} = a E\{X\} + b E\{Y\} + c$ appears as Corollary 4.2.2 of DeGroot and Schervish (p. 219) and contains this problem as the case $a = 2$, $b = 1$, $c = 7$; HPS invokes the same properties as “derived in Volume I” (p. 8) and passes expectations through linear combinations without comment thereafter (e.g. in Theorem 2 of §4.5, pp. 124–125). The final step substitutes $E\{X\} = 1$ and $E\{Y\} = 2$. None of the independence or covariance information was needed: linearity holds under arbitrary dependence, and the dependence structure first matters one moment up, at the variance. Pitman’s method of indicators (pp. 168–171) is the classic demonstration of this robustness—expectations of counts computed as sums of probabilities, however entangled the indicators may be. ■

Problem 14

Work out $\text{Var}\{Y + 3Z + 7\}$.

SOLUTION:

$$\begin{aligned}\text{Var}\{Y + 3Z + 7\} &= \text{Var}\{Y + 3Z\} && \text{(shift rule)} \\ &= \text{Var}\{Y\} + \text{Var}\{3Z\} + 2 \text{Cov}(Y, 3Z) && \text{(variance of a sum)} \\ &= \text{Var}\{Y\} + 9 \text{Var}\{Z\} + 6 \text{Cov}(Y, Z) && \text{(scaling rules)} \\ &= 9 + 9(16) + 6(-1) && \text{(substitute the givens)} \\ &= 9 + 144 - 6 = \boxed{147}.\end{aligned}$$

The opening step is the shift rule, $\text{Var}\{V + c\} = \text{Var}\{V\}$: adding a constant relocates the distribution without changing its spread, since $(V + c) - \text{E}\{V + c\} = V - \text{E}\{V\}$. Blitzstein and Hwang list it among the basic properties of variance and follow it with the warning that variance is not linear (pp. 172–173); Ross derives the combined shift-and-scale statement as “a useful identity” (p. 126); DeGroot and Schervish prove it as Theorem 4.3.4 (p. 229); Pitman states it on the standard-deviation scale, $\text{SD}(aX + b) = |a| \text{SD}(X)$ (p. 188). The second step is the variance of a sum, $\text{Var}\{U + V\} = \text{Var}\{U\} + \text{Var}\{V\} + 2 \text{Cov}(U, V)$, applied with $U = Y$ and $V = 3Z$ (Pitman, p. 430; Blitzstein and Hwang, p. 327; Ross gives the n -variable version as Eq. (4.1), p. 306, of which this is the case $n = 2$). The third step applies the scaling rules $\text{Var}\{aV\} = a^2 \text{Var}\{V\}$ and $\text{Cov}(U, aV) = a \text{Cov}(U, V)$, and the last substitutes the given values. As a check, Corollary 4.6.1 of DeGroot and Schervish (p. 253) delivers the closed form $\text{Var}\{aX + bY + c\} = a^2 \text{Var}\{X\} + b^2 \text{Var}\{Y\} + 2ab \text{Cov}(X, Y)$ in one stroke; applied

to the pair (Y, Z) with $a = 1$, $b = 3$, $c = 7$, it returns $9 + 144 - 6 = 147$ directly.

The dependence information enters exactly once, through $\text{Cov}(Y, Z) = -1$. Had Y and Z been independent, the cross term would vanish and the variances would simply add—the special case stated separately in every text (Pitman’s addition rule for variances, p. 193; DeGroot and Schervish, Thm. 4.3.5, p. 230), and the one variance rule HPS itself recalls from Volume I: “the variance of the sum of independent variables is equal to the sum of the variances” (p. 20). The independence of X and Y plays no role here, as X does not appear. ■

Problem 15

Work out $\text{Cov}(X + 4Y + 7, Y + 5Z + 7)$.

SOLUTION:

$$\begin{aligned}\text{Cov}(X + 4Y + 7, Y + 5Z + 7) &= \text{Cov}(X + 4Y, Y + 5Z) && \text{(shift rule)} \\ &= \text{Cov}(X, Y) + 5 \text{Cov}(X, Z) + 4 \text{Cov}(Y, Y) + 20 \text{Cov}(Y, Z) && \text{(bilinearity)} \\ &= 0 + 5 \text{Cov}(X, Z) + 4 \text{Var}\{Y\} + 20 \text{Cov}(Y, Z) && \text{(independence; covariance with itself)} \\ &= 5(1) + 4(9) + 20(-1) && \text{(substitute the givens)} \\ &= 5 + 36 - 20 = \boxed{21}.\end{aligned}$$

The constants disappear first: $\text{Cov}(U + a, V + b) = \text{Cov}(U, V)$, because covariance is computed from centered variables (DeGroot and Schervish leave the general form $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$ as Exercise 5 of §4.6, p. 255; the degenerate case $\text{Cov}(X, c) = 0$ appears in Blitzstein and Hwang's property list, p. 327). The expansion step is bilinearity: covariances of sums expand term by term, with scalars factoring out of each slot, exactly as in multiplying out $(x + 4y)(y + 5z)$ —

$$\text{Cov}(X + 4Y, Y + 5Z) = \text{Cov}(X, Y) + 5 \text{Cov}(X, Z) + 4 \text{Cov}(Y, Y) + 20 \text{Cov}(Y, Z).$$

Pitman states the general principle—the covariance of two linear combinations expands as the double sum of coefficient-weighted covariances (p. 444)—and Ross's Proposi-

tion 4.2 collects symmetry, scaling, and the double-sum expansion (§7.4, pp. 305–306). The third step uses two identities at once: independence of X and Y forces $\text{Cov}(X, Y) = 0$, which is Problem 12 (DeGroot and Schervish, Thm. 4.6.4, p. 251; Blitzstein and Hwang, Thm. 7.3.2, p. 327; Ross, p. 305; Pitman, p. 430), and a covariance with itself is a variance, $\text{Cov}(Y, Y) = \text{E}\{(Y - \mu_Y)^2\} = \text{Var}\{Y\}$ (Ross, Prop. 4.2(ii); DeGroot and Schervish, within the proof of Thm. 4.6.7, p. 254). Substituting $\text{Cov}(X, Z) = 1$, $\text{Var}\{Y\} = 9$, and $\text{Cov}(Y, Z) = -1$ completes the computation.

A last check across the three parts: each given value enters exactly where a rule calls for it, and two of them— $\text{E}\{Z\} = 3$ and $\text{Var}\{X\} = 4$ —are never needed at all. The rules, not the numbers, drive these computations. ■

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